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PROCESS FOR CONVERSION OF HYDROCRACKED PITCH

Abstract

The present disclosure relates to a process for conversion of hydrocracked pitch into distillates in a slurry hydro-conversion reactor in presence of hydrogen gas with recycling of a hydrocracked pitch residue to achieve improved overall conversion and minimize fresh addition of catalyst. The slurry hydro-conversion of hydrocracked vacuum residue along with recycle of hydrocracked pitch residue (4d) provides improved overall conversion of >97% with very less yield of <3 wt. % pitch (>540° C.) along with distillates (C6+–370° C.) in a yield of 55 wt. % and VGO (370-540° C.) in a yield of 28.3 wt. %. The recycling of hydrocracked pitch residue back to slurry hydro-conversion reactor facilitates the minimization of the addition of fresh catalyst to the reactor section.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to conversion of hydrocracked pitch into distillates in slurry hydro-conversion process.

BACKGROUND

[0002] Residue hydro-processing can be used to upgrade high boiling materials, such as petroleum residues including residues derived from extra heavy oils, pitch and oil sands bitumen by converting it into more valuable lighter distillates in presence of catalyst and hydrogen in different cracking reactors. Residues must be upgraded in a primary conversion unit before they can be further processed into marketable products. Primary upgrading units known in the prior art include, but are not restricted to, thermal cracking processes, such as visbreaking, delayed or fluidized coking, solvent deasphalting, and hydrogen addition processes such as fixed bed, moving bed, ebullated bed or slurry hydrocracking processes. It is evident from literature that hydrogen addition processes results in higher conversions of residues to distillates.

[0003] Ebullated bed reactor is one type of cracking reactor that utilizes ebullition, or bubbling, to achieve appropriate mixing of reactant feed and extrudate catalysts for residue upgradation and recycle gas bubbles up through the hydrocarbon mixture and the catalyst particles, creating a turbulent suspension. Solvent Deasphalting is another resid upgrading process where extraction process is used to remove the asphaltenes from the refinery residues to prepare a suitable feedstock for catalytic conversion unit. In deasphalting process, Paraffinic solvent is used as an extraction fluid for the removal of asphaltenes. The solvent deasphalting (SDA) unit is usually configured after the vacuum distillation tower. The output from the de-asphalting unit is de-asphalted oil (DAO) and asphalt pitch. The integration of above mentioned two processes results in enhancing conversion of residues to distillates.

[0004] Conventional integrated ebullated bed reactor hydrocracking with SDA process results 75-85% of conversion of virgin vacuum residue and result in high amount of hydrocracked residue as a pitch 15-25 wt % of the feed. U.S. Pat. No. 7,214,308 discloses an overall residue conversion of greater than 65%.

[0005] However, there is inevitably a volume of asphaltenic pitch that needs to be disposed of. SDA pitch features high viscosity and contains large amounts of components such as mainly asphaltenes that affect the atmospheric environment (sulfur, nitrogen, carbon residue, and heavy-metal components). Asphaltenes are the heaviest, most polar and structurally most complex components in heavy oils and residues. The pitch, if it is to be converted to fuel oil, requires a very high volume of a lighter cutter stock to be blended or can be alternatively fed to a Coking unit to produce coke, but pitch is not a very desirable feedstock as it contains poly-dispersed aromatics with high molecular weight, aromaticity and heteroatom content.

[0006] The slurry-phase hydrocracking is a promising process for converting residual feedstocks with high impurities such as atmospheric resid, vacuum resid, tar sands bitumen, heavy coker gas oils, partially hydrocracked heavy hydrocarbons, hydrocarbons that boil above 400 deg C. characterized by high Conradson Carbon Number, Asphaltenes, metals, sulfur, etc., such as but not limited to asphaltenic pitch from solvent extraction unit, delayed coker, mild resid hydrocracker, Pyrolysis oils, etc., to high value lower boiling products. In slurry-phase hydrocracking, the catalyst is in dispersed phase. The main functions of the dispersed catalyst are converting hydrogen gas into active hydrogen, suppressing coke formation, and promoting hydro-conversion. The liquid

feed stocks are mixed with hydrogen and dispersed catalyst particles, such as metal sulfides provide a high active surface area which actively participates in asphaltene hydrocracking reactions. Therefore, the pitch upgrading in slurry hydrocracking process would benefit the overall process by converting into lighter distillates there by reducing pitch yield.

OBJECTIVES

[0007] The present disclosure relates to a process for conversion of hydrocracked pitch into distillates in a slurry hydro-conversion reactor in presence of hydrogen gas with recycling of a hydrocracked pitch residue.

[0008] An objective of the present disclosure is to provide a process which has improved overall conversion.

[0009] Another objective of the present disclosure is to provide a process in which the addition of fresh catalyst has been minimized.

[0010] Yet another objective of the present disclosure is to provide a process with improved overall conversion of >97%.

[0011] Further an objective of the present disclosure is to provide a process with very less yield of <3 wt. % pitch (>540° C.) along with distillates (C6+–370° C.) and VGO (370-540° C.).

SUMMARY

[0012] The present disclosure relates to conversion of hydrocracked pitch into distillates in slurry hydro-conversion process. The present disclosure provides a process for conversion of hydrocracked pitch where a virgin vacuum residue feed stream (1) is hydrocracked in a hydro-conversion reactor (2) to obtain a hydrocracked effluent stream (2a). A hydrocarbon fraction (3a) and a hydrocracked vacuum resid fraction as bottom stream (4) is obtained by fractionating the hydrocracked effluent stream (2a) in a fractionator (3). Further, the hydrocracked vacuum resid fraction (4) is split into two portions viz., a first portion of hydrocracked stream (4a) and a second portion of hydrocracked stream (4b). The first portion of hydrocracked stream (4a) is sent to a Solvent De-asphalting unit (5) which provides a stream of asphaltenic pitch (8). A mixed feed stream (4c) is obtained by mixing the second portion of the hydrocracked stream (4b), a virgin vacuum residue (1a) and the asphaltenic pitch (8). The mixed stream (4c) is processed in a slurry hydro-conversion reactor (11) in presence of hydrogen (9) and a catalyst (10) to obtain an effluent stream (11a). A vapour effluent (12a) and a liquid effluent (12(b)) is obtained upon separation of the effluent stream (11a) in a separation zone (12). The liquid effluent 12(b) is fractionated to obtain lighter hydrocarbon fractions (13a, 13b, 13c) and unconverted hydrocracked pitch residue (14) which contains active metal sulfide catalyst particles along with hydrocarbon fractions. The unconverted hydrocracked pitch (14) is separated in a catalyst recovery zone (15) to obtain a purge stream (17) and a hydrocarbon stream (18) containing highly active metal sulfide catalyst particles with recovered hydrocarbons. The hydrocarbon stream (18) is recycled to the mixed stream (4c) to obtain a hydrocracked pitch residue (4d). The hydrocracked pitch residue (4d) is processed in the slurry hydro-conversion reactor (11).

[0013] The process as provided in the present disclosure can achieve enhanced residue conversions to >97-99% with asphaltenic pitch reduction to <1-3 wt % as a purge which needs to be utilized elsewhere by processing the hydrocracked vacuum residue in slurry reactor along with liquid recycle.

[0014] These and other features, aspects, and advantages of the present subject matter will become better understood with reference to the following description. This summary is provided to introduce a selection of concepts in a simplified form. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF ACCOMPANYING DRAWINGS

[0015] These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

[0016] FIG. 1 illustrate a schematic flow scheme showing the apparatus involved in the process for conversion of hydrocracked pitch.

DETAILED DESCRIPTION

[0017] The present disclosure relates to a process for conversion of hydrocracked pitch into distillates in a slurry hydro-conversion reactor in presence of hydrogen gas with recycling of a hydrocracked pitch residue. The schematic flow in FIG. 1 illustrates a process and apparatus for conversion of hydrocracked pitch into distillates in a slurry hydro-conversion reactor. FIG. 1 depicts the overall process of conversion of virgin vacuum residue feed stream (1) in a primary hydro-conversion unit such as ebullated bed reactors (2) resulting in light boiling materials along with unconverted high boiling hydrocarbons. As used herein, virgin residue, or other terms referring to residuum hydrocarbons, refers to hydrocarbon fractions having boiling points or a boiling range above about 540° C., but could also include hydrocarbon streams such as vacuum residuum, heavy deasphalted oil, hydrocracked atmospheric tower or vacuum tower bottoms resulted from extra heavy oils, tar sands bitumen, containing high-metal and impurity streams. The residuum described may be converted in a hydrocracking reaction system having one or more reaction stages including one or more hydrocracking reactors and different catalysts.

[0018] Following hydrocracking, the effluent (2a) from the primary conversion unit (2) may be fractionated in a fractionator (3) to recover one or more hydrocarbon fractions (3a). such as a light naphtha fraction, a heavy naphtha fraction, a kerosene fraction, a diesel fraction, a light vacuum gas oil fraction, a heavy gas oil fraction, and a hydrocracked vacuum resid fraction as bottom stream (4). This bottom stream (4) is split into a first portion of hydrocracked stream (4a) and a second portion of hydrocracked stream (4b). The first portion of hydrocracked stream (4a) is sent to a solvent deasphalting (SDA) unit (5) to produce a deasphalted oil (DAO) fraction and precipitated asphaltenic pitch (8). Deasphalted oil fraction may be processed in a DAO hydroconversion reactor (6) or sent to primary conversion unit (2) for improving distillate (7) yields.

[0019] The recycling of hydrocracked pitch residue back to slurry hydro-conversion reactor helps to minimize addition of fresh catalyst concentration to the reactor section, thereby significantly reducing the need to add fresh catalyst.

[0020] Second portion of the hydrocracked stream (4b) and the Asphaltenic pitch (8) mixed with and/or occasionally mixed with virgin vacuum residue (1a) to obtain a mixed feed stream (4c). Mixed feed stream (4c) in presence of hydrogen (9) and catalyst (10) will be further processed in a slurry hydroconversion reactor (11) to obtain an effluent stream (11a) to maximize conversions to distillate products and to reduce pitch yield from SDA unit. The mixed feed stream (4c) to slurry hydroconversion reactor is high severity in terms of impurities and high boiling material (>560° C.). The feed Conradson Carbon Residue (CCR) is in the range of 1-60 wt. %; more preferentially in the range of 5-40 wt. %. The asphaltene amount in the feed stream varies in the range of 1-60 wt. % and more preferentially in the range of 5-30 wt. %. Sulfur concentration in the mixed feed stream varies in the range of 0.5-7 wt. %; more preferably in the range of 1-6 wt. %.

[0021] The slurry hydroconversion reactor (11) may be of bubble column reactor, tubular reactor, fixed bed reactor, moving bed reactor, loop reactor or combination of any of these. Generally, the slurry hydro-conversion reaction process can include a reactor operating either in up-flow or down-flow. The reactor can be a bubble column slurry reactor or tubular plug flow reactor through which the feed, catalyst, and gas passes upwardly. Generally, the reaction temperature can be in the range of 380-500° C., more preferably about 420-460° C., and a hydrogen pressure of about 2-20 MPa,

more preferably in the range of 10-19 Mpa. The liquid hourly space velocity is typically in the range of 0.1-8 hr.sup.-1 and more preferably in the range of 0.5-6 hr.sup.-1. Hydrogen to hydrocarbon ratio of 100-5000 Litres/kg will be used and more preferably in the range of 500-2000 litre/kg. The catalyst concentration is in the range of 0.01-10 wt. % and more preferably in the range of 0.05-5 wt. %. The catalyst composition used is a metal-based oil soluble catalyst, solid dispersed nano particle supported catalyst, metal oxide based supports or a combination of thereof for the slurry hydro-conversion operation. Metal-based oil soluble catalyst comprises of mono metallic or bimetallic or trimetallic liquid catalyst in which metals comprising from group of molybdenum, nickel, cobalt, tungsten, iron metals and combination of them.

[0022] Liquid (12b) and vapor (12a) effluent from the slurry hydroconversion reactor (11) may be separated in HP & LP separator zone (12). This separation zone comprising of combination of high-pressure high-temperature (HPHT), high-pressure Low-temperature (HPLT), Low-pressure high-temperature (LPHT), and Low-pressure low-temperature (LPLT) separator(s) for separating liquids with catalyst from the vapours. The hydrogen containing gas stream (12a) then be routed through a gas cooling, purification, and recycle gas compression system (not shown) to recycle the hydrogen. The separated liquid streams (12b) further sent to product fractionation column (13) to separate naphtha (13a), distillates kerosene and diesel (13b), vacuum gas oil (13c) and unconverted hydrocracked pitch residue (14). This unconverted hydrocracked pitch residue contains catalyst metal sulfide particles which are very active along with hydrocarbons boiling between 520-650° C. will be recovered in catalyst recovery zone (15). Diluents (16) such as clarified oil, VGO, and other aromatic solvents may be added to stream (14) to remove high dense materials including hard to convert asphaltenes along with metals and coke precursors if any as a purge stream (17) and to obtain a hydrocarbon stream (18) containing highly active metal sulfide catalyst particles with recovered hydrocarbons.

[0023] The metal sulfide catalyst particles are recovered from catalyst recovery zone (15). The catalyst recovery zone (15) may comprise centrifugal separator, settler for phase separation, vacuum separator, and filtration setup to remove metals and coke precursors. Further, the stream (18) containing highly active metal sulfide particles with recovered hydrocarbons are mixed with mixed feed stream (4c) to obtain a hydrocracked pitch residue (4d) which is recycled back to slurry hydro-conversion reactor to minimize addition of fresh catalyst concentration to the reactor section. By processing the hydrocracked vacuum residue in slurry reactor along with liquid recycle can achieve enhanced residue conversions to >97-99% with asphaltenic pitch reduction to <1-3 wt % as a purge which needs to be utilized elsewhere.

[0024] The present disclosure is further illustrated by reference to the following examples which is for illustrative purpose only and does not limit the scope of the disclosure in any way. These examples are not intended to be inclusive of all aspects of the subject matter disclosed herein, but rather to illustrate representative features, methods, compositions, and results. These examples are not intended to exclude equivalents and variations of the present disclosure, which are apparent to one skilled in the art.

Example 1

[0025] The present disclosure is tested in a pilot scale slurry hydroconversion unit. Hydrocracked residue as a feed tested in this experiment and properties are density of 1.064 g/cc, Sulphur of 5.5 wt %, CCR of 34%, and Asphaltene content of 22 wt %. The oil soluble molybdenum-based catalyst used in this example is 0.1% to the feed and the reaction conducted at 430° C., 17 Mpa, and 0.25 h.sup.-1 LHSV. The slurry hydro-conversion of hydrocracked residue results are shown in Table 1.

TABLE-US-00001 TABLE 1 Yield, wt. % Product Present Disclosure Exemplary Data Drygas 4.8 3.3 LPG 4.8 2.9 H2S 4.7 1.3 Distillates (C6+-370° C.) 54.5 42.5 VGO (370-540° C.) 28.3 26.9 Pitch (540+° C.) 2.8 23.1 Overall Conversion, % 97.2 77

[0026] Conventional integrated ebullated bed reactor hydrocracking with SDA process results 75-

85% of conversion of virgin vacuum residue and result in high amount of hydrocracked residue as a pitch 15-25 wt. % of the feed. In the present disclosure, the slurry hydro-conversion of hydrocracked residue showed improved overall conversion of >97% with very less yield of <3 wt. % pitch (540+° C.) which is to be removed as purge stream. The process as mentioned in the present disclosure provides distillates (C6+–370° C.) with a yield of 55 wt. % and VGO (370-540° C.) with yield of 28.3 wt. %. As per the metal analysis of unconverted heavy hydrocarbon (540+° C.) from the process of the present disclosure metal sulfide particle content in the range of 0.09-0.1% which is mainly concentrated in heavier fraction.

Example 2

[0027] The present disclosure is tested in a pilot scale slurry hydro-conversion unit. Hydrocracked residue as a feed tested in this experiment and properties are CCR of 40%, and Asphaltene content of 30 wt. %. The oil soluble molybdenum based catalyst used in this example is 0.5% to the feed and the reaction conducted at 430° C., 17 Mpa, and 0.25 h.sup.–1 LHSV. The results are showed in below Table 2.

TABLE-US-00002 TABLE 2 Product Yield, wt. % Drygas 6.30 LPG 1.70 H2S 1.10 Distillates (C6+–370° C.) 29.8 VGO (370-540° C.) 35.0 Pitch (540+° C.) 16.1 Overall Conversion, % 83.9

Example 3

[0028] The present disclosure is tested in a pilot scale slurry hydroconversion unit. Hydrocracked residue as a feed tested in this experiment and properties are density of 1.064 g/cc, Sulphur of 5.5 wt. %, CCR of 34%, and Asphaltene content of 22 wt. %. The oil soluble molybdenum-based catalyst used in this example is 0.2% to the feed and the reaction conducted at 430° C., 17 Mpa, and 0.33 h.sup.–1 LHSV. The slurry hydro-conversion of hydrocracked residue results are shown in Table 3.

TABLE-US-00003 TABLE 3 Product Yield, wt. % Drygas 3.8 LPG 3.0 H2S 4.0 Distillates (C6+–370° C.) 50.4 VGO (370-540° C.) 32.3 Pitch (540+° C.) 6.5 Overall Conversion, % 93.5

[0029] From the results, it is concluded that the slurry hydro-conversion process is very promising to integrate with existing low severity resid processing units in order to achieve higher conversions and distillate fuels.

ADVANTAGES

[0030] 1. Improved overall residue conversion of >97%. [0031] 2. Minimization of need to add fresh catalyst to a process for conversion of hydrocracked pitch. [0032] 3. Very less yield of <3 wt. % pitch (>540° C.).

[0033] Although the subject matter has been described in considerable detail with reference to certain preferred aspects thereof, other aspects are possible. As such, the spirit and scope of the subject matter should not be limited to the description of the preferred aspects contained therein.

Claims

1. A process for conversion of hydrocracked pitch, said process comprising: a) hydrocracking a virgin vacuum residue feed stream in a hydro-conversion reactor to obtain a hydrocracked effluent stream; b) fractionating the hydrocracked effluent stream in a fractionator to obtain a hydrocarbon fraction and a hydrocracked vacuum resid fraction as bottom stream; c) splitting the hydrocracked vacuum resid fraction into a first portion of hydrocracked stream and a second portion of hydrocracked stream; d) sending the first portion of hydrocracked stream to a Solvent De-asphalting unit which provides a stream of asphaltenic pitch; e) mixing the second portion of the hydrocracked stream, a virgin vacuum residue and the asphaltenic pitch to obtain a mixed feed stream; f) processing the mixed stream in presence of hydrogen and a catalyst in a slurry hydro-conversion reactor to obtain an effluent stream; g) separating the effluent stream in a separation zone, to obtain a vapour effluent and a liquid effluent; h) fractionating the liquid effluent in a fractionator to obtain lighter hydrocarbon fractions and unconverted hydrocracked pitch residue

which contains active metal sulfide catalyst particles along with hydrocarbon fractions; i) separating the unconverted hydrocracked pitch in a catalyst recovery zone to obtain a purge stream and a hydrocarbon stream containing highly active metal sulfide catalyst particles with recovered hydrocarbons; j) recycling the hydrocarbon stream to the mixed stream to obtain a hydrocracked pitch residue; and k) processing the hydrocracked pitch residue in the slurry hydro-conversion reactor.

2. The process as claimed in claim 1, wherein the virgin vacuum residue feed stream is selected from a group consisting of hydrocarbon fractions having boiling points above 540° C., heavy deasphalted oil, extra heavy oils, tar sands bitumen, containing high-metal and impurity streams resulting as bottom streams from hydrocracked atmospheric tower or vacuum tower.

3. The process as claimed in claim 1, wherein the hydro-conversion reactor is selected from a group consisting of a fixed bed reactor, a moving bed hydroconversion unit, and an ebullated bed hydroconversion unit, or combination thereof.

4. The process as claimed in claim 3, wherein the hydro-conversion unit can co-exist with a thermal cracking unit, or a solvent extraction unit.

5. The process as claimed in claim 1, wherein the hydrocarbon fraction comprises a light naphtha fraction, a heavy naphtha fraction, a kerosene fraction, a diesel fraction, a light vacuum gas oil fraction, and a heavy gas oil fraction.

6. The process as claimed in claim 1, wherein the slurry hydro-conversion reactor is selected from a group consisting of a bubble column reactor, a tubular reactor, a fixed bed reactor and a loop reactor.

7. The process as claimed in claim 5, wherein the slurry hydro-conversion reactor comprises of an assembly of multiple reactors.

8. The process as claimed in claim 1, wherein the slurry hydro-conversion reactor is operated at a temperature in the range of 380-500° C., at a hydrogen pressure in the range of 2-20 MPa, and at a liquid hourly space velocity in the range of 0.1-8 hr.sup.-1.

9. The process as claimed in claim 7, wherein the slurry hydro-conversion reactor is operated at a temperature in the range of 420-460° C., at a hydrogen pressure in the range of 10-19 MPa, and at a liquid hourly space velocity in the range of 0.5-6 hr.sup.-1.

10. The process as claimed in claim 1, wherein the catalyst is present in a concentration in the range of 0.01-10 wt. %.

11. The process as claimed in claim 1, wherein the catalyst is selected from a group consisting of a metal based oil soluble catalyst, solid dispersed nano particle supported catalyst, and metal oxide based supports or a combination of thereof.

12. The process as claimed in claim 9, wherein the metal based oil soluble catalyst comprises mono metallic or bimetallic or trimetallic liquid catalyst with metal selected from group consisting of molybdenum, nickel, cobalt, and tungsten or combination thereof.

13. The process as claimed in claim 1, wherein the separation zone comprises of separators, selected from a group of high pressure high temperature (HPHT) separator, high pressure Low temperature (HPLT) separator, Low pressure high temperature (LPHT) separator, and Low pressure low temperature (LPLT) separator for separating liquids with catalyst from the vapors or liquids.

14. The process as claimed in claim 1, wherein the hydrocracked pitch residue (**4d**) processed in slurry hydro-conversion reactor has asphaltene content in the range of 1-70 wt. %.

15. The process as claimed in claim 13, wherein the hydrocracked pitch residue processed in slurry hydro-conversion reactor has asphaltene content in the range of 5-30 wt. %.

16. The process as claimed in claim 1, wherein the hydrocracked pitch residue has a Conradson Carbon residue in the range of 1-60 wt. %.

17. The process as claimed in claim 15, wherein the hydrocracked pitch residue has a Conradson Carbon residue in the range of 5-40 wt. %.

18. The process as claimed in claim 1, wherein the catalyst recovery section is selected from a

group comprising of a centrifugal separator, a settler for phase separation, a vacuum separator, and a filtration setup.
